HiGUM: A Hierarchical Gaussian Model for Cooperative Multi-UAV Smoke Monitoring

David Saldaña
Computer Science and Engineering
Lehigh University
Bethlehem, USA
saldana@lehigh.edu

Yunyao Li
Earth and Environmental Sciences
University of Texas at Arlington
Arlington, USA
yunyao.li@uta.edu

Diego Patiño
Computer Science and Engineering
University of Texas at Arlington
Arlington, USA
diego.patino@uta.edu

***Abstract*—Monitoring wildland-fire smoke with unmanned aerial vehicles (UAVs) requires representing large three-dimensional datasets of smoke concentration while retaining enough spatial details for cooperative planning. This paper presents a hierarchical Gaussian model (HGM) that can serve as a compact, multi-resolution representation of a static scalar smoke concentration point cloud. The HGM recursively partitions the dataset using two-component Gaussian mixture models and stores, at each tree node, a Gaussian approximation of the point distribution and the mean $PM_{2.5}$ concentration. A selected hierarchy level supplies component centroids as candidate waypoints for a UAV fleet. These waypoints are allocated using k-means, ordered through Hamiltonian cycles, and tracked by fixed-wing vehicles using nonlinear model predictive control. We demonstrate the complete pipeline on point cloud data extracted from a realistic smoke simulation, visualize the smoke plume using up to 32 Gaussian ellipsoids, generate dynamically feasible plans for fleets of three and five UAVs, and evaluate HGM construction across 15 levels using 257,896 smoke samples. Our results show that the hierarchy connects compact plume representation with selectable planning resolution and independently computable vehicle trajectories.**

***Index Terms*—Multi-Robot Systems, UAV, Probabilistic Models, Environmental Monitoring**

I. Introduction

Measuring the location, extent, and concentration of wildland-fire smoke is essential for understanding population exposure and supporting timely air-quality decisions. Fire smoke is a major source of fine particulate matter such as $PM_{2.5}$ and reactive gases. Exposure to $PM_{2.5}$ is associated with respiratory diseases and increased mortality. Landscape-fire smoke is estimated to cause approximately 339,000 premature deaths annually worldwide [1], [2]. Because a plume can extend far beyond the active fire, useful monitoring requires spatially resolved measurements of the three-dimensional concentration field, not only observations at the fire perimeter.

Existing methods provide complementary but incomplete views of this field. Satellite observations cover large areas but provide low resolution information. Numerical models support air-quality forecasting and exceedance prediction [3], [4], but their results remain sensitive to real fire emissions. Large field campaigns, including FIREX-AQ [5] and WE-CAN [6], combine aircraft, ground, and remote-sensing observations to obtain detailed plume measurements, but such campaigns provide limited spatial and temporal coverage and are difficult to deploy routinely. Robotic sensing offers a complementary approach: prior work uses single- and multi-robot systems to observe, track, and predict environmental boundaries [7]–[9]. However, efficiently coordinating a UAV fleet over a large volumetric smoke space requires a representation of the scene that is both compact and compatible with multi-resolution planning.

This requirement exposes a gap between plume data prod-

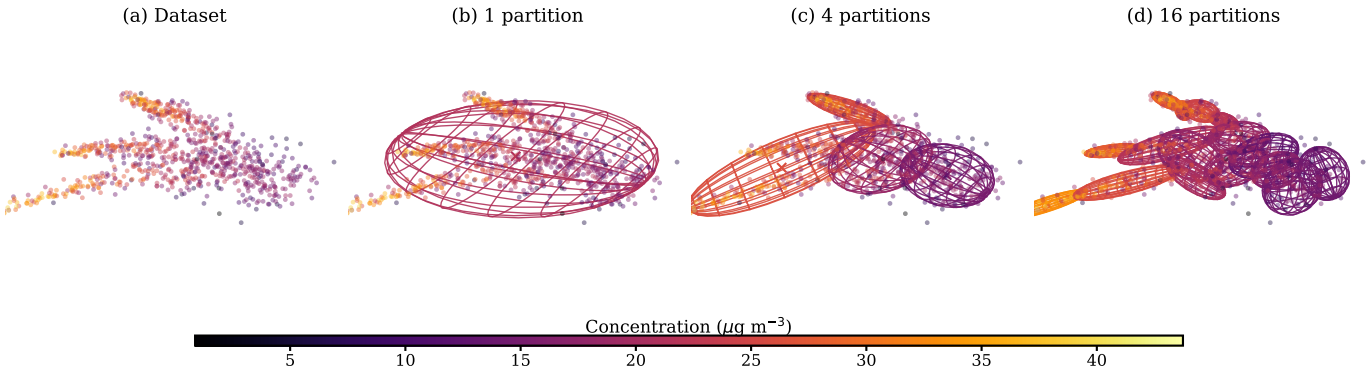


Fig. 1. Hierarchical representation of a synthetic combined three-source smoke plume: (a) dataset point-cloud, (b) one Gaussian representing all samples, (c) four leaf partitions, and (d) sixteen leaf partitions. The three source centerlines begin at distinct crosswind locations, and rise and converge toward a common downwind trajectory. Each ellipsoid shows the two-standard-deviation surface of its Gaussian and is colored by the average concentration of its dataset subset.

ucts and robotic planning. A dense point cloud or simulation grid preserves spatial detail, but using every sample as a planning target is computationally expensive. Conversely, a single summary distribution is compact but cannot represent the irregular, non-convex geometry of a smoke plume. Existing Gaussian representations demonstrate that complex three-dimensional geometries can be encoded and rendered efficiently using collections of Gaussian components [10], [11]. However, they do not directly provide a hierarchical abstraction of the smoke plume, nor are they utilized for multi-UAV planning pipelines as considered here. We address this gap by leveraging a hierarchical Gaussian model (HGM) that organizes a smoke-concentration point cloud into a tree of increasingly refined Gaussian components. Each node stores a Gaussian approximation of the spatial distribution of its points and their mean scalar $PM_{2.5}$ concentration. A planner selects a tree level based on the desired resolution, assigns the resulting component centroids to the UAVs, computes an ordered cycle for each vehicle, and tracks the waypoints using model predictive control (MPC). The formulation applies to point clouds obtained from UAV measurements, satellite or in-situ observations, or atmospheric simulations.

The main contributions of this paper are threefold: (1) A compact, tree-structured Gaussian representation of a scalar three-dimensional smoke-concentration dataset at multiple resolutions; (2) a cooperative planning method that uses the Gaussian tree hierarchy for waypoint selection and allocation, Hamiltonian-cycle ordering, and fixed-wing MPC trajectory generation; and (3) an evaluation on WRF-FIRE-derived smoke plume data demonstrating the HGM representation at different refinement levels, multi-UAV trajectories, and the computational cost of constructing the hierarchy.

II. Hierarchical Gaussian Model

Our model approximates the smoke concentration field in a 3D region around a wildland-fire as a structured mixture of Gaussian components organized across multiple spatial scales, we call hierarchical Gaussian model (HGM). We represent the hierarchy as a tree data structure, such as an Octree [12], [13], with the root node containing a single 3D Gaussian distribution that summarizes all collected measurements across the environment.

The HGM is then a tree-structured mixture of Gaussian components that captures the spatial distribution of smoke at multiple resolutions. Each node in the tree corresponds to a Gaussian component, and its hierarchical structure enables adaptive refinement in regions of interest, such as those with high concentration gradients or high uncertainty.

Let $\Omega \subset \mathbb{R}^3$ denote the monitored airspace containing a smoke plume. The plume induces a scalar concentration field

$$c : \Omega \times \mathbb{R}_{\geq 0} \to \mathbb{R}_{\geq 0}, \qquad (\mathbf{p}, t) \mapsto c(\mathbf{p}, t), \tag{1}$$

where $\mathbf{p} = [x, y, z]^\mathsf{T} \in \Omega$ consists of the coordinates of a spatial location, $t \in \mathbb{R}_{\geq 0}$ represents time, and $c(\mathbf{p}, t)$ is the smoke concentration given a specific location and time. For a snapshot acquired at time t_0, we write $c(\mathbf{p}) \equiv c(\mathbf{p}, t_0)$.

We assume that we observe the smoke plume through a concentration point cloud of N samples,

$$\mathcal{D} = \{(\mathbf{p}_i, c_i)\}_{i=1}^{N}, \qquad \mathbf{p}_i \in \Omega, \quad c_i \in \mathbb{R}_{\geq 0}, \tag{2}$$

where $\mathbf{p}_i$ is the location of sample i, and c_i is its measured or estimated smoke concentration. The dataset $\mathcal{D}$ may be obtained from UAV measurements or by combining heterogeneous data sources, including WRF–FIRE simulations, satellite products, and other in-situ sensors. Regardless of its source, each sample is represented by the same position–concentration pair. The objective of the hierarchical Gaussian model is to construct a compact, multi-resolution approximation $\widehat{c}(\mathbf{p})$ of the field $c(\mathbf{p})$ from the observed point cloud $\mathcal{D}$.

At the coarsest resolution, we represent the entire dataset $\mathcal{D}$ using the root tuple in the HGM tree, such that

$$\mathcal{R}_r = (\boldsymbol{\mu}_r, \boldsymbol{\Sigma}_r, \bar{c}_r), \tag{3}$$

where $\boldsymbol{\mu}_r \in \mathbb{R}^3$ is the point cloud's centroid and $\boldsymbol{\Sigma}_r \in \mathbb{R}^{3\times 3}$ is its covariance matrix. The covariance matrix $\boldsymbol{\Sigma}$ represents the Gaussian orientation and extent, and can be visualized as an ellipsoid whose principal directions are the eigenvectors of $\boldsymbol{\Sigma}_r$ and whose squared principal-axis lengths are proportional to its eigenvalues. Together, $(\boldsymbol{\mu}_r, \boldsymbol{\Sigma}_r)$ parameterize the Gaussian distribution of all point locations in $\mathcal{D}$, while $\bar{c}_r$ is their average concentration. A constant-density surface of this Gaussian is the ellipsoid $(\mathbf{p}-\boldsymbol{\mu}_r)^\mathsf{T}\boldsymbol{\Sigma}_r^{-1}(\mathbf{p}-\boldsymbol{\mu}_r) = \gamma$, where $\gamma > 0$ selects the displayed confidence level. This representation is highly compact, since it requires only three parameters for the mean, six independent parameters for the covariance matrix, and one scalar value, $\bar{c}_r$, for the representative concentration. However, a single ellipsoid cannot accurately represent a plume with a curved, multi-modal, filamentary, or otherwise non-Gaussian shape.

To increase fidelity, a Gaussian associated with a node is recursively split and replaced by a mixture of k child Gaussians. In principle, k may be selected independently at each node to minimize tree size subject to a prescribed approximation error. In this paper, we set $k = 2$ for every split. The resulting hierarchy is therefore a binary tree, which simplifies construction and traversal time while retaining a coarse-to-fine representation of the plume.

A. Hierarchical Tree Structure

We organize $\mathcal{D}$ using the rooted binary tree $\mathcal{T} = (\mathcal{V}, \mathcal{E}, r)$, where $\mathcal{V}$ and $\mathcal{E}$ are the node and edge sets, respectively, with r the root node. For a node $v \in \mathcal{V}$, let $\ell(v)$ denote its depth, $\mathrm{pa}(v)$ its parent, and $\mathrm{ch}(v)$ its children. Each node is associated with a subset $\mathcal{D}_v \subseteq \mathcal{D}$ containing the samples assigned to that node. The root contains the complete dataset, i.e., $\mathcal{D}_r = \mathcal{D}$. The children nodes partition their parent dataset such that

$$\mathcal{D}_v = \bigcup_{u \in \mathrm{ch}(v)} \mathcal{D}_u, \qquad \mathcal{D}_u \cap \mathcal{D}_{u'} = \varnothing \quad \text{if} \quad (u \neq u'). \tag{4}$$

Every internal node has exactly two children, $|\,\mathrm{ch}(v)| = 2$, whereas every leaf has no children.

Every node v has the same tuple representation

$$\mathcal{R}_v = (\boldsymbol{\mu}_v, \boldsymbol{\Sigma}_v, \bar{c}_v), \tag{5}$$

where $\boldsymbol{\mu}_v \in \mathbb{R}^3$ and $\boldsymbol{\Sigma}_v \in \mathbb{R}^{3\times3}$ are, respectively, the spatial mean and three-dimensional covariance matrix of the points assigned to the node. Smoke concentration is not a parameter of the Gaussian; instead, $\bar{c}_v$ is the representative scalar concentration of those points. The concentration in the node's tuple is an ordinary arithmetic mean $\bar{c}_v = \frac{1}{N_v}\sum_{(\mathbf{p}_i,c_i)\in\mathcal{D}_v} c_i$, where $N_v = |\mathcal{D}_v|$ is the number of points assigned to node v. Thus, $\mathcal{R}_r$ represents the complete dataset.

B. Partitioning the dataset

Starting with the root, we recursively partition each node in the tree into multiple 3D Gaussian distributions using a Gaussian Mixture Model via Expectation Maximization [14], [15]. This is useful when a single Gaussian cannot capture the local plume shape, i.e., the GMM identifies spatially coherent groups of points, allowing each child to represent a simpler local structure with its own mean and covariance. For the binary tree, $\mathrm{ch}(v) = \{v_1, v_2\}$. We fit a two-component, full-covariance GMM to only the spatial coordinates $\{\mathbf{p}_i : (\mathbf{p}_i, c_i) \in \mathcal{D}_v\}$; the concentration measurements do not affect the partition.

Expectation–Maximization (EM) is an iterative algorithm for estimating parameters in probabilistic models that contain latent, or unobserved, variables. In our case, the latent variable is the unknown child-component assignment of each point in $\mathcal{D}_v$. The expectation step estimates the probability of each possible assignment, while the maximization step uses these probabilities to update the mixture weights, means, and covariance matrices of the children nodes. For each child $u \in \mathrm{ch}(v)$, we define a Gaussian component, and define $\alpha_{u|v} \in [0,1]$ as the probability that a point drawn from $\mathcal{D}_v$ belongs to component u, and therefore represents the expected fraction of the parent's points assigned to that child. The weights satisfy $\sum_{u\in\mathrm{ch}(v)} \alpha_{u|v} = 1$ and may be initialized uniformly, i.e., $\alpha_{u|v} = 1/2$ for a binary split.

We denote by $p(\mathbf{p}_i \mid u)$ the conditional spatial likelihood of point $\mathbf{p}_i$ under child u. This likelihood is the three-dimensional Gaussian density parameterized by $\boldsymbol{\mu}_u$ and $\boldsymbol{\Sigma}_u$. The responsibility $r_{iu} = P(u \mid \mathbf{p}_i)$ is the posterior probability that point $\mathbf{p}_i$ belongs to child component u, conditioned on its position and the current GMM parameters. Given current parameters, the expectation step computes the responsibility of child u for point i as

$$r_{iu} = \frac{\alpha_{u|v}\, p(\mathbf{p}_i \mid u)}{\sum_{w\in\{v_1,v_2\}} \alpha_{w|v}\, p(\mathbf{p}_i \mid w)}. \tag{6}$$

After EM converges, the point is assigned to the child with the largest responsibility to obtain the disjoint datasets used by the tree.

The maximization step updates the component parameters following

$$\begin{aligned} N_u^{(r)} &= \sum_{(\mathbf{p}_i,c_i)\in\mathcal{D}_v} r_{iu}, & \alpha_{u|v} &= \frac{N_u^{(r)}}{N_v}, \\ \boldsymbol{\mu}_u &= \frac{1}{N_u^{(r)}} \sum_{(\mathbf{p}_i,c_i)\in\mathcal{D}_v} r_{iu}\mathbf{p}_i, \\ \boldsymbol{\delta}_{iu} &= \mathbf{p}_i - \boldsymbol{\mu}_u, \\ \boldsymbol{\Sigma}_u &= \frac{1}{N_u^{(r)}} \sum_{(\mathbf{p}_i,c_i)\in\mathcal{D}_v} r_{iu}\boldsymbol{\delta}_{iu}\boldsymbol{\delta}_{iu}^{\mathsf{T}} + \epsilon\mathbf{I}. \end{aligned} \tag{7}$$

We repeat the expectation and maximization steps until the log-likelihood converges [14], [15]. We then form a disjoint hard partition by assigning each point to its maximum-responsibility component,

$$\mathcal{D}_u = \left\{ (\mathbf{p}_i, c_i) \in \mathcal{D}_v : u = \operatorname*{arg\,max}_{w\in\{v_1,v_2\}} r_{iw} \right\}. \tag{8}$$

We subsequently compute each child tuple $\mathcal{R}_u = (\boldsymbol{\mu}_u, \boldsymbol{\Sigma}_u, \bar{c}_u)$ from its assigned subset, with $\bar{c}_u$ given by the ordinary average. A split is accepted only when both children contain at least the prescribed minimum number of points; otherwise, v remains a leaf. Figure 2 illustrates the resulting coarse-to-fine representation on the synthetic plume.

To evaluate our approach on a a more complex topology, we consider a synthetic dataset obtained by combining three initially separate smoke sources whose plumes merge downwind, as depicted in Fig. 1.

An additional advantage of the tree structure is parallelism. Once a node has been split, the datasets associated with nodes on different branches are disjoint. Consequently, fitting the subsequent GMM and computing their child partitions requires no shared data and can be computed concurrently. This branch-level parallelism reduces hierarchy-construction time and supports multi-core and distributed implementations.

This paper considers only the hierarchical representation of a static dataset. Nevertheless, we can extend the same model to incorporate new measurements without recomputing the complete hierarchy from scratch. In particular, only the nodes affected by a new sample need to be updated.

In a distributed implementation, UAV j maintains a local subtree $\mathcal{T}_j \subseteq \mathcal{T}$ and communicates only changed sufficient statistics. Because independent statistics are additive, observations collected by different UAVs can be fused by summing their increments before updating the affected node tuple.

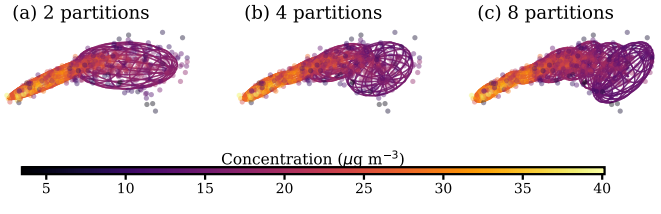


Fig. 2. Recursive binary GMM partitioning of the synthetic plume: (a) 2, (b) 4, and (c) 8 leaf partitions. At each level, our method splits each parent dataset into a two-component GMM fitted to their spatial coordinates.

III. Cooperative Planning with Multiple UAVs

Multiple UAVs enable rapid monitoring of smoke plumes, but their effectiveness depends on coordinating the regions visited by each vehicle [7]. We use the HGM centroids as compact points of interest, allocate these points among the vehicles, and employ model predictive control (MPC) to track the resulting waypoint sequences.

To address the high computational cost of planning over several points of interest [16], we exploit the hierarchical structure of the HGM. At coarse levels, the model enables fast allocation of UAVs to informative regions, reducing the global planning complexity. Each UAV then computes an independent trajectory within its assigned subregion. During execution, the UAV can decide to re-plan online based on local measurements and explore deeper levels of the hierarchy. To enable real-time computation, we select a level of the HGM whose nodes define a set of ellipsoids with centroids $\boldsymbol{\mu}_v$ (see Fig. 2). The horizontal projection $\mathbf{q}_v = [\mu_{v,x}, \mu_{v,y}]^\mathsf{T}$ of each centroid defines a candidate waypoint.

We consider a fleet of n fixed-wing UAVs, indexed by $i \in \{1, \dots, n\}$. We apply k-means [17] to the waypoint coordinates with the number of clusters set to n. This clustering produces a disjoint set $\widetilde{\mathcal{Q}}_i = \{\widetilde{\mathbf{q}}_{i,1}, \dots, \widetilde{\mathbf{q}}_{i,M_i}\}$ containing the M_i projected HGM centroids assigned to UAV i. The cycle is rotated so that its first waypoint is the one closest to the initial vehicle position. Finding the shortest Hamiltonian cycle is NP-hard. For the small instances considered here, e.g., $M_i < 100$, we can use an exact solver in practice. For larger waypoint sets, the same pipeline can use an approximate or heuristic tour without changing the MPC formulation.

Let $m_i \in \{1, \dots, M_i\}$ be the index of the waypoint currently pursued by UAV i. We define the active MPC reference as

$$\mathbf{q}_i^\star = \mathbf{q}_{i,m_i}.$$

After applying an MPC control step, the waypoint index advances when the horizontal position $\mathbf{p}^{(i)} = [x^{(i)}, y^{(i)}]^\mathsf{T}$ is within the prescribed tolerance $\varepsilon_q > 0$:

$$m_i^+ = \begin{cases} \min(m_i + 1, M_i), & \|\mathbf{p}^{(i)} - \mathbf{q}_{i,m_i}\|_2 < \varepsilon_q, \\ m_i, & \text{otherwise.} \end{cases}$$

Thus, we recompute $\mathbf{q}_i^\star$ by selecting the next assigned HGM centroid rather than by solving an additional continuous optimization problem. For simplicity, the vehicles move at a prescribed constant altitude $\bar{z} \in \mathbb{R}$. In the inertial Cartesian frame, x and y are the horizontal coordinates of a plane parallel to the ground plane, and z is the coordinate of the vertical axis. Hence, $z_t^{(i)} = \bar{z}$ for every vehicle and time step. Let $x_t^{(i)}$ and $y_t^{(i)}$ denote the horizontal coordinates of the ith vehicle, and let $\psi_t^{(i)}$ denote its yaw angle at time step t. Its three-dimensional position is therefore $\mathbf{r}_t^{(i)} = [x_t^{(i)}, y_t^{(i)}, \bar{z}]^\mathsf{T}$, while the state used by the planner is

$$\mathbf{x}_t^{(i)} = \begin{bmatrix} x_t^{(i)} & y_t^{(i)} & \psi_t^{(i)} \end{bmatrix}^\mathsf{T}.$$

Roll and pitch are not included because the planner uses a reduced-order model and the lower-level flight controller tracks the resulting commands. The forward speed v_0 is fixed, and the control input is the scalar steering command $u_t^{(i)} = \delta_t^{(i)}$. Using an Ackermann-type planar model with effective turning-length parameter L, the forward-Euler dynamics are

$$\begin{aligned} x_{t+1}^{(i)} &= x_t^{(i)} + \Delta t\, v_0 \cos \psi_t^{(i)}, \\ y_{t+1}^{(i)} &= y_t^{(i)} + \Delta t\, v_0 \sin \psi_t^{(i)}, \\ \psi_{t+1}^{(i)} &= \psi_t^{(i)} + \Delta t\, \frac{v_0}{L} \tan \delta_t^{(i)}. \end{aligned}$$

We can extend this formulation to variable-altitude flight by adding altitude to the state and vertical speed to the control input.

For an MPC horizon of N steps, let $\delta_{-1}^{(i)}$ denote the steering command applied immediately before the current optimization. The implemented objective is

$$J_i = \sum_{t=0}^{N-1} \left[w_p \left\| \mathbf{p}_t^{(i)} - \mathbf{q}_i^\star \right\|_2^2 + w_u \left(\delta_t^{(i)} \right)^2 \right],$$

where $\mathbf{p}_t^{(i)} = [x_t^{(i)}, y_t^{(i)}]^\mathsf{T}$. The two terms penalize waypoint-tracking error, and steering magnitude. Steering is constrained by $|\delta_t^{(i)}| \le \delta_{\max}$. We solve the resulting nonlinear program with the Interior Point OPTimizer (IPOPT), a primal–dual interior-point method with a filter line-search strategy for large-scale nonlinear programming [18], in a receding-horizon loop. Once the active waypoint is reached within a prescribed tolerance, the controller advances to the next waypoint in $\mathcal{Q}_i$. Because the waypoint sets are disjoint, the trajectory of each UAV can be computed independently.

IV. Experiments

Our proposed framework is validated through a series of simulations that demonstrate the feasibility of the hierarchical Gaussian model for multi-UAV plume monitoring. We evaluate the model's ability to represent a complex plume structure, scalability with the number of levels in the hierarchy and the number of UAVs.

We implement the prototype in Python, and use use NumPy to load and manipulate the WRF-FIRE volumetric data, construct the point clouds, and compute node statistics. We implement the full-covariance GMMs and their EM estimation using scikit-learn. The nonlinear MPC problem is formulated symbolically with CasADi and solved through its IPOPT interface.

a) Testing scenario: We have developed an initial end-to-end prototype of the pipeline that integrates WRF-FIRE simulations with a probabilistic model for multi-UAV planning. As a test scenario, we use a wildfire simulation over Colorado, USA, whose WRF-FIRE output in NetCDF format provides smoke concentrations on a georeferenced grid at irregular altitudes. We pre-process this NetCDF file by correcting the altitude coordinates and interpolating the concentration field onto a regular three-dimensional grid. We use the (x, y, z) coordinates associated with each grid's cell center to obtain the point-cloud representation shown in Fig. 3(a). The HGM represents the same dataset at different resolutions: level 2

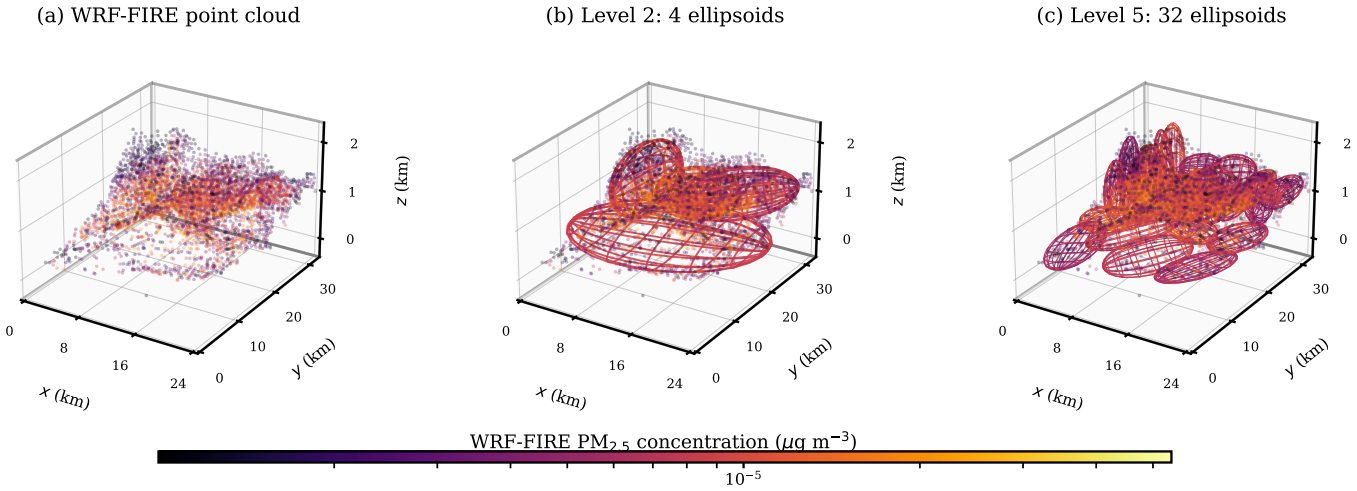


Fig. 3. WRF-FIRE smoke dataset and its hierarchical Gaussian representation. (a) We extract the point cloud from simulation time index 3 by retaining voxels with smoke concentration at least 10^{-6}, taking every 50th valid voxel, and mapping the volumetric-array indices to (x, y, z) in kilometers using the 100 m grid spacing. (b) HGM level 2 represents the same dataset using four Gaussian ellipsoids. (c) HGM level 5 represents it using 32 ellipsoids. Every parent dataset is recursively partitioned by a two-component, full-covariance GMM, and each wireframe shows the two-standard-deviation surface of its node Gaussian. Point and ellipsoid colors indicate the WRF-FIRE $PM_{2.5}$ concentration in $\mu g\,m^{-3}$ on a logarithmic scale.

contains four Gaussian ellipsoids [Fig. 3(b)], whereas level 5 contains 32 ellipsoids [Fig. 3(c)]. This hierarchy provides adaptive resolution while balancing representation detail and computational cost.

b) Generating optimal trajectories: Building on this representation, we compute UAV trajectories at levels 3 and 4 for a team of three UAVs and at level 5 for a team of five UAVs (Fig. 4). At each level, k-means assigns the HGM centroids to the vehicles, and an exact depot-aware Hamiltonian-cycle solution determines the visiting sequence within each cluster. For each ordered sequence, the receding-horizon MPC repeatedly solves the constrained optimal-control problem and applies its first control input, producing the dynamically feasible trajectories shown in the figure. These results demonstrate the feasibility of combining model-driven representation with scalable, dynamics-aware planning for multi-UAV plume monitoring.

c) UAV scalability: To evaluate the computational scalability of our representation, we measure the cumulative wall-clock time required to construct the HGM from the root through levels 1–15. The benchmark uses all 257,896 valid WRF-FIRE smoke voxels at simulation time index 3, rather than the subsampled point cloud used only for visualization. Figure 5 reports the mean of five single-threaded runs. The measurements include the recursive two-component, full-covariance GMM fits at every splittable node. We retain singleton nodes as terminal leaves because a two-component GMM is undefined for a single sample. Absolute wall-clock values depend on the computing platform, but the trend characterizes the scaling of the current Python implementation.

V. Conclusion and Future Work

This paper presented a hierarchical Gaussian model for compact, multi-resolution representation of three-dimensional smoke-concentration point clouds and for cooperative UAV planning over those representations. The method offers a binary hierarchy whose levels determine the tradeoff between representation compactness and plume-shape detail. This representation provides waypoints for multiple UAVs, and an MPC controller generates trajectories that respect the reduced-order fixed-wing dynamics. Experiments using a WRF-FIRE smoke field demonstrated the complete pipeline at multiple HGM resolutions and fleet sizes. Our results show that the HGM provides a practical interface between volumetric plume data and multi-UAV trajectory generation, while allowing the representation resolution to be selected based on available computational resources and mission requirements.

Future work includes temporal and velocity parameters for dynamic plumes, online assimilation of UAV measurements, collision and communication constraints, heuristic Hamiltonian-cycle solvers for large waypoint sets, and validation against satellite and in-situ observations.

Acknowledgment

The authors gratefully acknowledge the support of the National Science Foundation (NSF). Awards 2322840 and 2442475.

References

[1] W. E. Cascio, "Wildland fire smoke and human health," *Science of The Total Environment*, vol. 624, pp. 586–595, 2018.

[2] F. H. Johnston, S. B. Henderson, Y. Chen, J. T. Randerson, M. Marlier, R. S. DeFries, P. Kinney, D. M. Bowman, and M. Brauer, "Estimated global mortality attributable to smoke from landscape fires," *Environmental Health Perspectives*, vol. 120, no. 5, pp. 695–701, 2012.

[3] Y. Li, D. Tong, S. Ma, S. R. Freitas, R. Ahmadov, M. Sofiev, X. Zhang, S. Kondragunta, R. Kahn, Y. Tang, B. Baker, P. Campbell, R. Saylor, G. Grell, and F. Li, "Impacts of estimated plume rise on $pm_{2.5}$ exceedance prediction during extreme wildfire events: a comparison of three schemes (briggs, freitas, and sofiev)," *Atmospheric Chemistry and Physics*, vol. 23, no. 5, pp. 3083–3101, 2023.

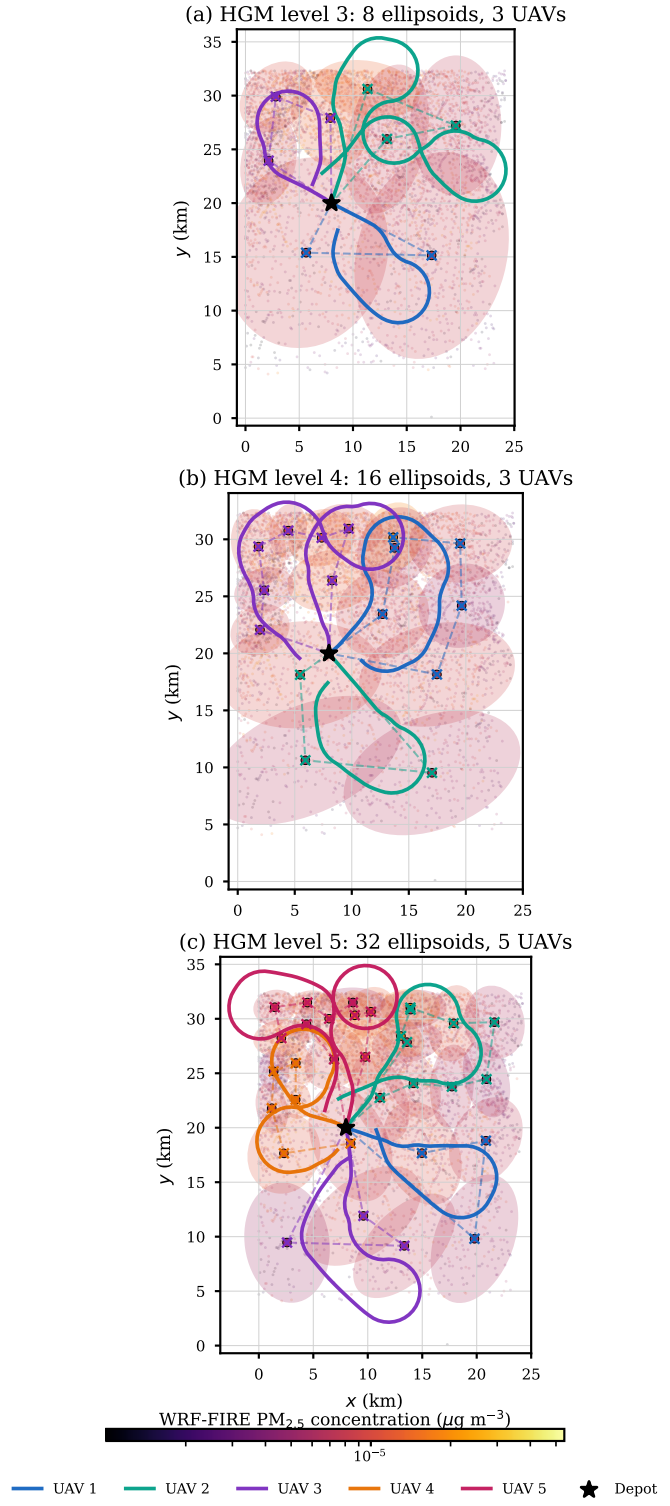


Fig. 4. Top-view plans over the HGM at different leves and number of UAVs. The thin dashed lines show these Hamiltonian cycles, while the solid colored curves are the trajectories generated by the MPC from the common depot (black star).

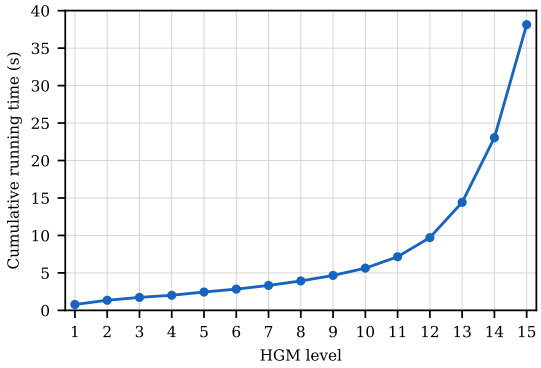


Fig. 5. Cumulative HGM construction time for levels 1–15 using the full WRF-FIRE point cloud of 257,896 valid voxels.

[4] Y. Li, D. Tong, P. Makkaroon, T. DelSole, Y. Tang, P. Campbell, B. Baker, M. Cohen, A. Darmenov, R. Ahmadov, E. James, E. Hyer, and P. Xian, "Multiagency ensemble forecast of wildfire air quality in the united states: Toward community consensus of early warning," *Bulletin of the American Meteorological Society*, vol. 105, no. 6, pp. E991 – E1003, 2024.

[5] C. Warneke, J. P. Schwarz, J. Dibb, O. Kalashnikova, G. Frost, J. Al-Saad, S. S. Brown, W. A. Brewer, A. Soja, F. C. Seidel *et al.*, "Fire influence on regional to global environments and air quality (firex-aq)," *Journal of Geophysical Research: Atmospheres*, vol. 128, no. 2, p. e2022JD037758, 2023.

[6] A. P. Sullivan, R. P. Pokhrel, Y. Shen, S. M. Murphy, D. W. Toohey, T. Campos, J. Lindaas, E. V. Fischer, and J. L. Collett Jr, "Examination of brown carbon absorption from wildfires in the western us during the we-can study," *Atmospheric Chemistry and Physics*, vol. 22, no. 20, pp. 13 389–13 406, 2022.

[7] D. Saldana, R. Assunçao, M. A. Hsieh, M. F. Campos, and V. Kumar, "Estimating boundary dynamics using robotic sensor networks with pointwise measurements," *Autonomous Robots*, vol. 45, no. 2, pp. 193–208, 2021.

[8] D. Saldaña, R. Assunção, M. A. Hsieh, M. F. M. Campos, and V. Kumar, "Cooperative prediction of time-varying boundaries with a team of robots," in *2017 International Symposium on Multi-Robot and Multi-Agent Systems (MRS)*, Dec 2017, pp. 9–16.

[9] D. Saldaña, R. J. Alitappeh, L. C. A. Pimenta, R. Assunção, and M. F. M. Campos, "Dynamic perimeter surveillance with a team of robots," in *2016 IEEE International Conference on Robotics and Automation (ICRA)*, May 2016, pp. 5289–5294.

[10] B. Kerbl, G. Kopanas, T. Leimkuehler, and G. Drettakis, "3d gaussian splatting for real-time radiance field rendering," *ACM Transactions on Graphics (TOG)*, vol. 42, pp. 1 – 14, 2023.

[11] A. Guédon and V. Lepetit, "Sugar: Surface-aligned gaussian splatting for efficient 3d mesh reconstruction and high-quality mesh rendering," *2024 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 5354–5363, 2023.

[12] D. J. Meagher, "Octree encoding: A new technique for the representation, manipulation and display of arbitrary 3-d objects by computer," in *Technical Report, Rensselaer Polytechnic Institute*, 1980, presents one of the first formal descriptions of the octree data structure for hierarchical 3D spatial subdivision and solid-object modeling.

[13] J. L. Bentley, "Multidimensional binary search trees used for associative searching," *Commun. ACM*, vol. 18, no. 9, p. 509–517, Sep. 1975.

[14] C. M. Bishop, *Pattern Recognition and Machine Learning*. Springer, 2006.

[15] K. Yamashita, S. Nobuhara, and K. Nishino, "3d-gmnet: Single-view 3d shape recovery as a gaussian mixture," *Proceedings of the British Machine Vision Conference 2020*, 2020.

[16] A. Bircher, M. Kamel, K. Alexis, H. Oleynikova, and R. Siegwart, "Receding horizon" next-best-view" planner for 3d exploration," in *2016 IEEE international conference on robotics and automation (ICRA)*. IEEE, 2016, pp. 1462–1468.

[17] S. P. Lloyd, "Least squares quantization in PCM," *IEEE Transactions on Information Theory*, vol. 28, no. 2, pp. 129–137, 1982.

[18] A. Wächter and L. T. Biegler, "On the implementation of an interior-point filter line-search algorithm for large-scale nonlinear programming," *Mathematical Programming*, vol. 106, no. 1, pp. 25–57, 2006.